Campus GHG Inventory and 2040 Net Zero Roadmap

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Objective and contribution

The project aimed to establish a greenhouse gas (GHG) baseline, identify major emission sources, and develop a staged reduction plan for the University of Seoul. I led the eight-member team and analyzed 46 commuting-survey responses to estimate emissions across student, faculty, and staff groups.

Inventory construction and findings

We selected 2023 as the baseline because pandemic conditions affected the preceding two years. Facility records, procurement information, and campus observations supplied activity inputs; Korean and IPCC emission factors converted these inputs into CO_2 equivalents. For commuting, travel distances and annual attendance were combined with mode-specific emission factors and a round-trip adjustment. Group averages were then scaled to the corresponding campus populations, with precision limited by the small survey sample.

Emission source	Activity input	2023 estimate (tCO_2e)
Purchased electricity	Facilities electricity-use records	10,373.5
Commuting	46 survey responses and population scaling	1,956.1
Stationary combustion	Facilities fuel-use records	1,763.0

Table 1. Selected estimated campus emission sources in 2023.

The report estimated net emissions of 14,868 tCO_2e for 2023 after including forest uptake. Purchased electricity accounted for about 70% of this estimate, making electricity supply and demand central to the proposed roadmap.

Roadmap and project outcome

We proposed a 50% reduction from the 2023 baseline by 2030 and net zero by 2040. The roadmap placed initial renewable generation, storage, and participation incentives in 2027, capacity expansion in 2030–2035, and vehicle-to-grid integration in 2040. Microgrid development would coordinate electricity supply and demand, while electricity sales were considered to finance carbon-credit purchases for residual emissions. Carbon capture, utilization, and storage was excluded because installation was judged difficult on campus.

The course roadmap linked energy infrastructure with waste management, participation incentives, and financing options. Projected reductions remain conditional on equipment-performance assumptions. The project ranked first among eight teams.